DIGITAL ELECTRONICS

ET 8301

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Lecture 03

BASIC LOGIC GATES

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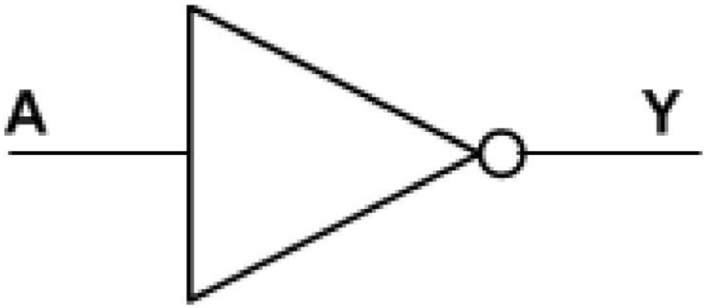
Introduction

- Logic gates are electronic circuits that can be used to ***implement*** the most elementary logic expressions, also known as ***Boolean expressions***.
- The logic gate is the most basic building block of ***combinational logic***.
- There are three basic logic gates, namely the **OR** gate, the **AND** gate and the **NOT** gate.
- Other logic gates that are derived from these basic gates are the **NAND** gate, the **NOR** gate, the **EXCLUSIVEOR** gate and the **EXCLUSIVE-NOR** gate.

Inverter

- The inverter performs the Boolean **NOT (complement) operation**.
- Standard logic symbols for inverter is



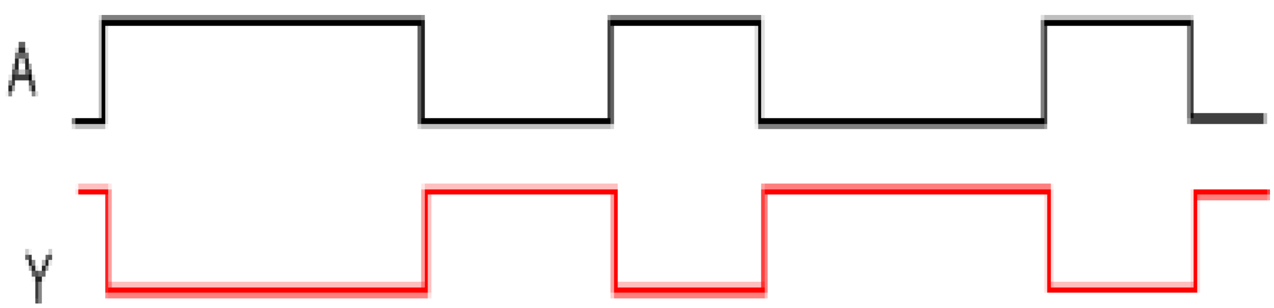
- The bubble indicate inversion or complementation.
- When the input is **LOW**, the output is **HIGH**; when the input is **HIGH**, the output is **LOW**.

Inverter

- Truth table of an Inverter

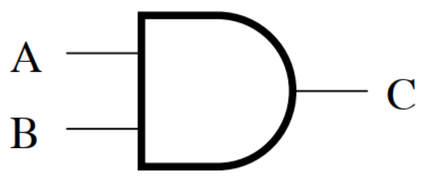
INPUT	OUTPUT
0	1
1	0

- Thus, the Boolean expression for an inverter is $Y = \overline{A}$
- Inverter waveforms



AND Gate

- An AND gate have ***two or more*** inputs and ***performs logical multiplication***.
- The AND function, which is also called ***logic product***, is represented by a **dot (·)**.
- Standard logic symbols for AND gate is



- The AND gate produces a **HIGH output** when all inputs are **HIGH**; otherwise, the output is **LOW**.

AND Gate

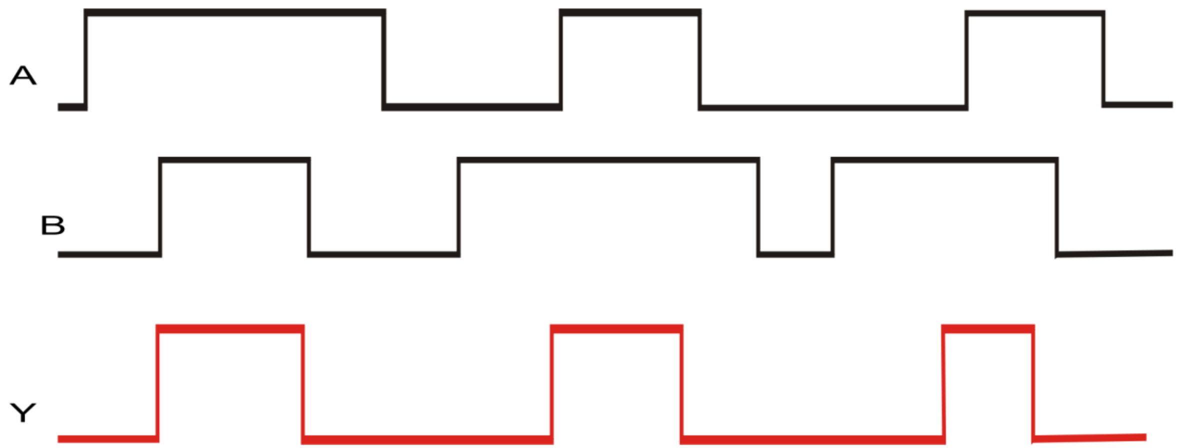
- For a 2-input gate, the truth table is

INPUT		OUTPUT
A	B	Y=A.B
0	0	0
0	1	0
1	0	0
1	1	1

- The AND operation is usually shown with a ***dot between the variables*** but it may be implied (no dot).Thus, the AND operation is written as **Y= A.B** or **Y = AB.**

AND Gate

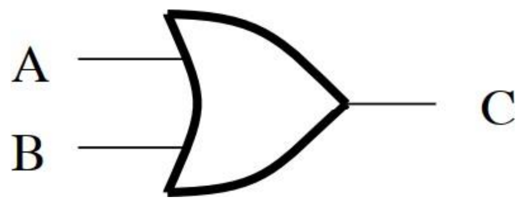
- Waveform for AND gate.



- The AND operation is used in ***computer programming*** as a selective mask. If you want to retain certain bits of a binary number but reset the other bits to 0, you could set a mask with 1's in the position of the retained bits.

OR Gate

- OR gate have two or more inputs and performs **logical addition** and is represented by a **plus (+)**.
- Standard logic symbols for OR gate is



OR $C=A+B$

- The OR gate produces a **HIGH output** if any input is **HIGH**; if all inputs are **LOW**, the output is **LOW.**

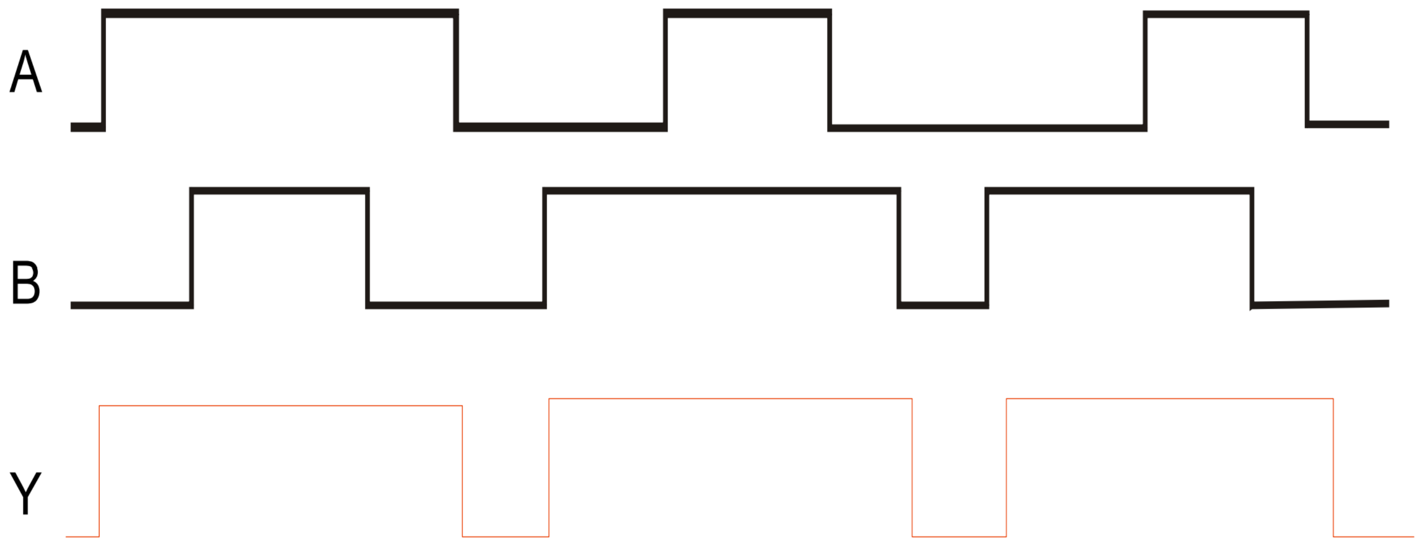
OR Gate

- For a 2-input gate, the truth table is

INPUT		OUTPUT
A	B	Y=A+B
0	0	0
0	1	1
1	0	1
1	1	1

OR Gate

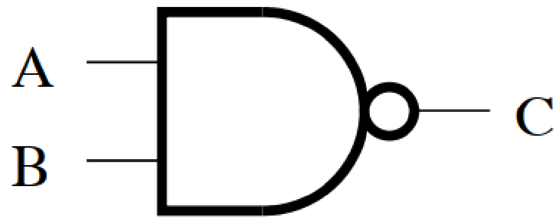
- The Waveform for a 2-input gate is



- The OR operation can be used in computer programming to set certain bits of a binary number to 1.

NAND Gate

- It is universal gate can be used to perform the **AND**, **OR** and **Inverter** operations.
- Standard logic symbols for **NAND** gate is



$$C = \overline{A \cdot B}$$

- The NAND gate produces a **LOW** output when all inputs are **HIGH**; otherwise, the output is **HIGH**.

NAND Gate

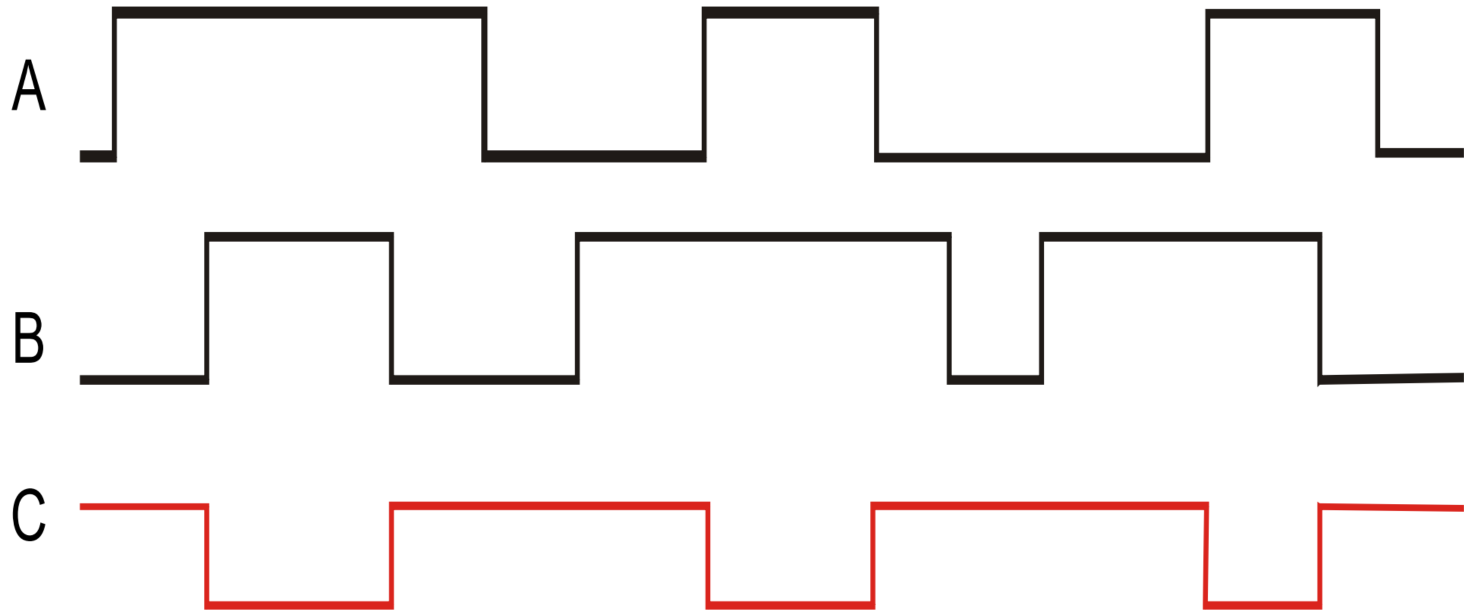
- For a 2-input gate, the truth table is

INPUT		OUTPUT
A	B	Y=$\overline{A.B}$
0	0	1
0	1	1
1	0	1
1	1	0

- The NAND operation is shown with a dot between the variables and an overbar covering them. Thus, the NAND operation is written as **Y**=$\overline{\boldsymbol{A}.\boldsymbol{B}}$

NAND Gate

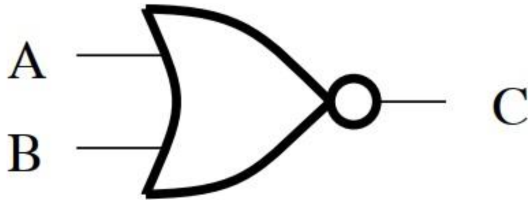
- Waveform



- The NAND gate is particularly useful because it is a "**universal**" gate – all other basic gates can be constructed from NAND gates.

NOR Gate

- It is also an universal gate, since **NOR** gates can be combined to perform **Inverter**, **AND** and **OR** operations.
- Standard logic symbols for NOR gate is



$$C = \overline{A + B}$$

- The **NOR gate** produces a **LOW** output if any input is **HIGH**; if all inputs are **HIGH**, the output is **LOW**.

NOR Gate

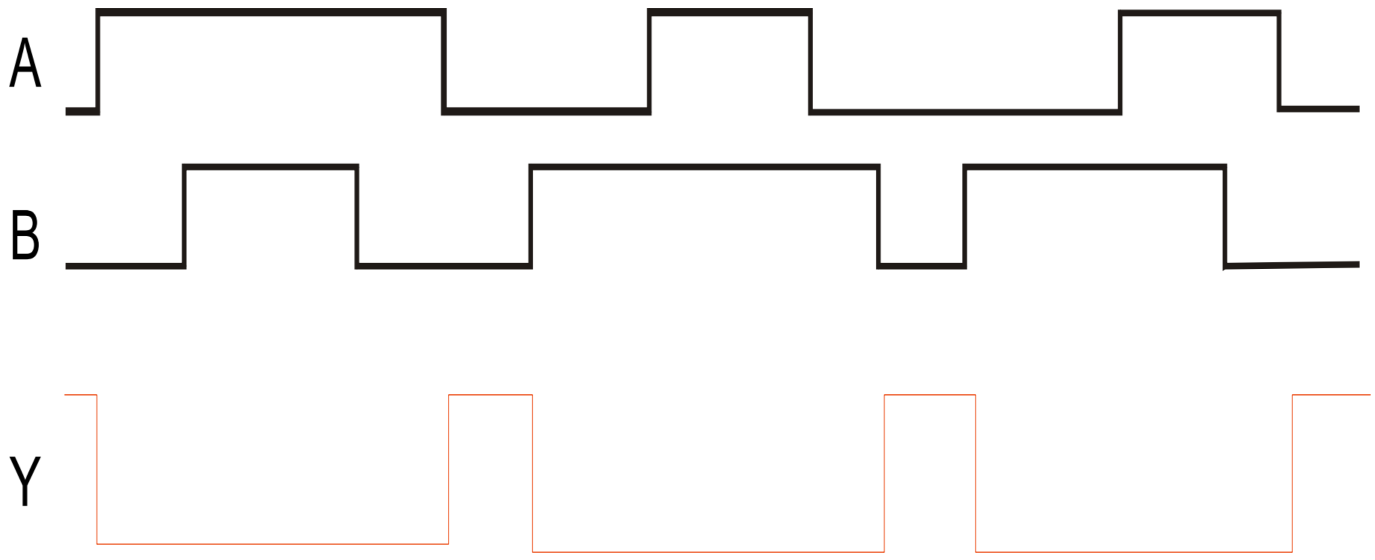
- For a 2-input gate, the truth table is

INPUTS		OUTPUTS
A	B	Y=$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

- The NOR operation is shown with a **plus sign (+)** between the variables and an **overbar covering** them. Thus, the NOR operation is written as $\boldsymbol{Y = \overline{A + B}}$**.**

NOR Gate

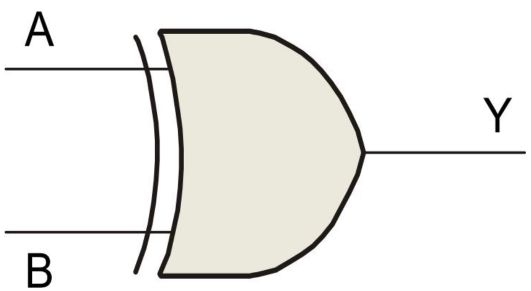
- Waveforms



- The NOR operation will produce a **LOW** if any input is **HIGH**.
- The NAND and NOR gates are considered to be **universal gates**. This means that any logic function can be implemented using just NAND or NOR gates.

XOR Gate

- It is formed by combining other gates.
- Standard logic symbols for XOR gate is



- The XOR gate produces a **HIGH** output only when both inputs are **at opposite logic** levels.

XOR Gate

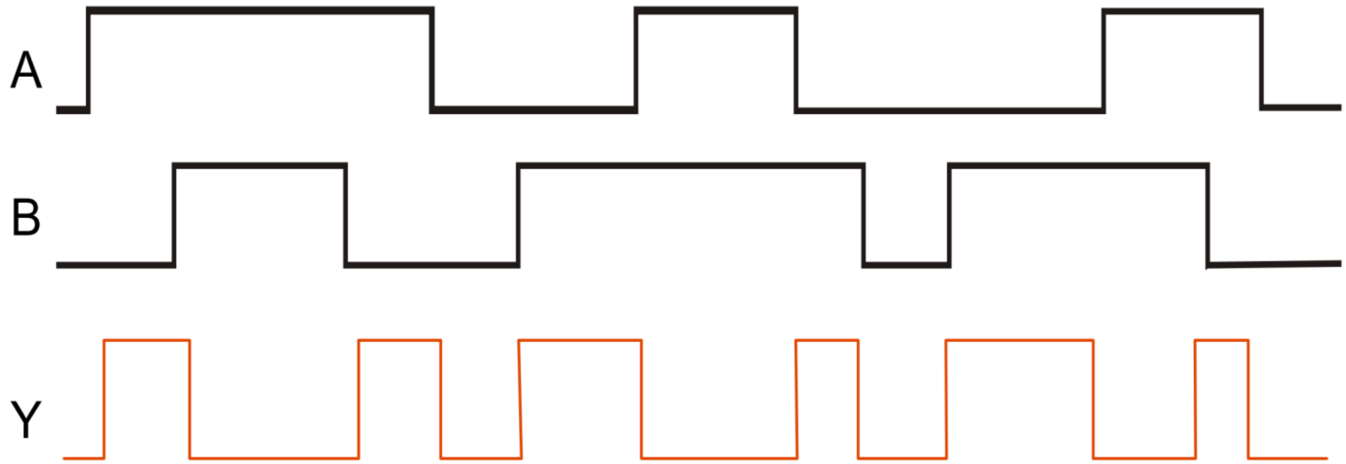
- For a 2-input gate, the truth table is

INPUT		OUTPUT
A	B	Y= $\bar{A}B + A\bar{B}$
0	0	0
0	1	1
1	0	1
1	1	0

- The XOR operation is written as **Y=**$AB + AB$.Alternatively, it can be written with a circled plus sign between the variables as $\boldsymbol{Y = A \oplus B}$

XOR Gate

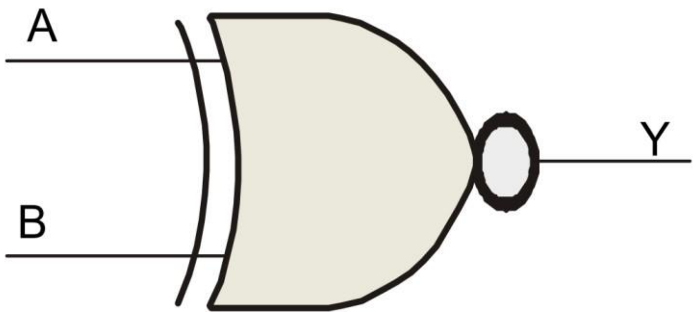
- Waveform



- Exclusive-OR gate can be used as a two bit adder. From XOR Truth table it can be seen that the output is the binary sum of the two input. When the inputs are both 1's the sum is 0 without carry

XNOR Gate

- Standard logic symbols for XNOR gate is



- The XNOR gate produces a **HIGH** output only when both ***inputs are at the same logic level***.

XNOR Gate

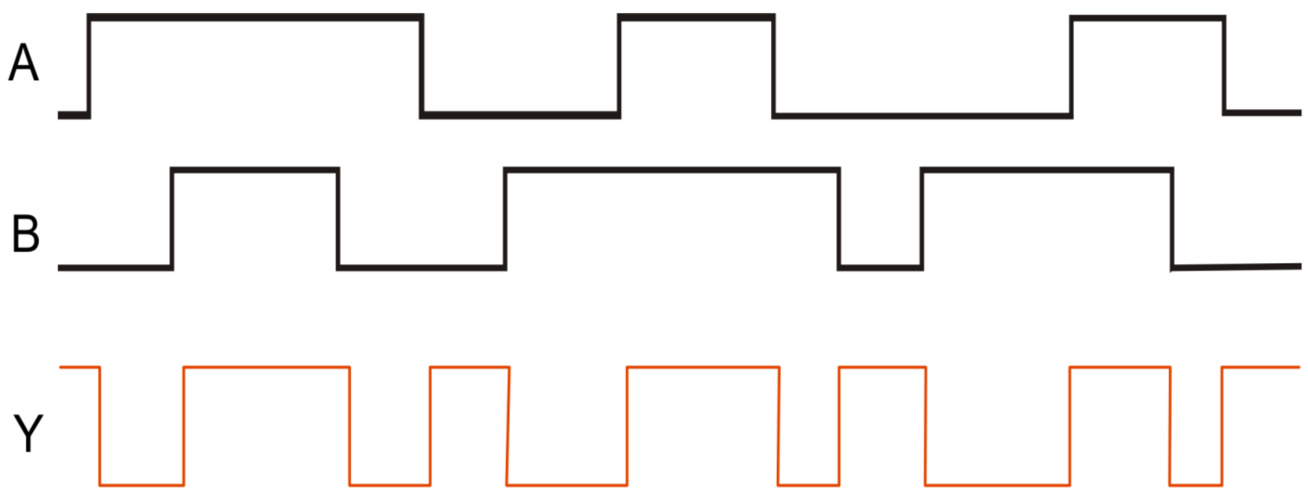
- The truth table for XNOR gate

INPUT		OUTPUT
A	B	Y = $\bar{A}\bar{B} + AB$
0	0	1
0	1	0
1	0	0
1	1	1

- The XNOR operation shown as $\boldsymbol{Y = \bar{A}\bar{B} + AB}$ Alternatively, the XNOR operation can be shown with a circled dot between the variables. Thus, it can be shown as $\boldsymbol{Y = A \odot B}$

XNOR Gate

- The waveform for XNOR gate



NB:

The NAND and NOR gates are considered to be universal gates. This means that any logic function can be implemented using just NAND or NOR gates. It must be noted that neither the XOR gate nor the XNOR gate is universal.

HOME WORK

- Short Notes about **Three-state buffer**

Next Lecture

BOOLEAN ALGEBRA